

From DenseNet+LSTM to CLIP+GRPO

An Iterative Research Study in Small-Scale Image Captioning

This report documents a complete improvement campaign for an image captioning system on Flickr8K (8,091 images). Across five model generations, three failed directions, a production deployment crisis, and an out-of-distribution failure analysis, BLEU-4 improved 41.8% and BLEU-1 improved 22.9% over baseline while serving at ~420 ms per caption. All numbers are measured on the same held-out split with identical evaluation code; negative results are reported with the same rigor as positive ones.

BLEU-1	BLEU-4	ROUGE-L	Latency
0.5334 → 0.6559	0.1218 → 0.1727	0.2216 → 0.2782	31.6 s → 0.42 s

1. Introduction and Scope

Image captioning on small datasets poses a specific challenge: modern captioners derive fluency from web-scale pretraining that a single-GPU project cannot replicate. The central research question: how far can systematic engineering — feature transfer, spatial attention, subword tokenization, beam search, and reinforcement learning — push a small decoder trained on 8k images, and where is the hard ceiling?

Methodology invariants. Every generation is trained and evaluated identically: Flickr8K split 6,877 train / 1,214 validation images, seed 42, NLTK sentence-BLEU (smoothing method-1), ROUGE-L via LCS dynamic programming, CIDEr-D with document frequencies fixed over training references (validated against pycocoevalcap before every RL run), and beam search with GNMT length normalization ($\alpha = 1.2$) plus a soft 2-gram repeat penalty (log 0.4).

2. Dataset and Evaluation Protocol

Property	Value
Images	8,091 Flickr8K photographs
References	5 crowd-sourced captions per image
Split	6,877 train / 1,214 validation (fixed, seed 42)
Tokenizer Gen 1	Keras word-level · vocab 8,427 · max len 35
Tokenizer Gen 2+	Byte-Pair Encoding 6k merges · max len 38
Preprocessing	lowercase, strip non-alphabetic, startseq/endseq wrap

2.1 Metric definitions

- **BLEU-n:** modified n-gram precision with method-1 smoothing.
- **ROUGE-L:** longest-common-subsequence F-measure vs the first reference.
- **CIDEr-D:** TF-IDF-weighted n-gram consensus ($\sigma = 6$); IDF fixed once over training references for RL stability.
- **Word precision:** share of generated words appearing in any reference (audit subsets, seed 7).
- **CHAIR-lite:** hallucinated nouns absent from the image's reference noun set.

3. Generation 1 — DenseNet201 + LSTM Baseline

3.1 Architecture

Frozen DenseNet201 (ImageNet) → global average pooling → single 1920-d vector. Decoder: Embedding(8427, 256) ⊕ projected image vector → LSTM(256) → Dropout(0.5) → residual add → Dense(128) → softmax. Trained with label smoothing 0.1; best epoch 26, val loss 4.7517.

3.2 Results

Decode	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L
Greedy	0.4372	—	—	—	—
Beam-10 α 1.2	0.5334	0.3203	0.1890	0.1218	0.2216

Key finding

Two structural flaws: global pooling erases WHERE objects are, and ImageNet features encode classification, not language grounding. Both motivated Generation 2.

4. Generation 2 — Spatial Attention v2

4.1 Architecture

Encoder frozen before norm5 → full 7×7×1920 grid flattened to 49 × 1920. Decoder: LSTM-512 with Bahdanau additive attention over all 49 positions; BPE-6k subword tokenizer. Recipe: AdamW lr 3e-4 / wd 0.01, label smoothing 0.05, embedding dropout 0.3, scheduled sampling ramp 0→0.25 (epochs 5–30), grad clip 5.0. Best epoch 24, val loss 3.8161.

4.2 Results (full split)

Model	Decode	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	CIDEr-D
Gen 1 baseline	beam-10	0.5334	0.3203	0.1890	0.1218	0.2216	—
Gen 2 CE	greedy	0.5840	0.3664	0.2268	0.1487	0.2507	—
Gen 2 CE	beam-5 α 1.2	0.5649	0.3704	0.2447	0.1663	0.2548	0.5397

Key finding

Spatial attention lifted BLEU-4 from 0.1218 to 0.1663 (+37%) with the SAME frozen encoder — the decoder was the first bottleneck, not the features. Calibration analysis: ECE 0.0119 (near-perfectly calibrated), top-1 accuracy 34.6% vs mean confidence 33.5%.

4.3 Beam sweep (17 configurations)

Config	BLEU-1	BLEU-4	ms/image
beam-10 α 1.2 ★	0.5358	0.1253	35.9
beam-10 α 1.0	0.5345	0.1249	36.8
beam-5 α 1.0	0.5347	0.1224	15.8
beam-7 α 1.2	0.5325	0.1223	23.7

beam-5 α 1.2 chosen for production: statistically tied quality, 2.3× faster than beam-10.

5. Generation 2-RL — GRPO on CIDEr-D

GRPO samples G=5 captions per image, rewards each by CIDEr-D, and reinforces above-group-mean samples (advantage = (r – mean)/std). lr 1e-5, 16 images/step, max length 24, EMA(0.999) weights served. Fast fixed-DF CIDEr scorer validated against pycocoevalcap before training.

Metric	Pre-RL	Post-RL (5 ep)	Δ
Val CIDEr-D (400 subset)	0.4165	0.4591	+10.2%
Mean train reward	0.157	0.287	+83%
Val BLEU-1 (subset)	0.5787	0.6013	+2.3 pts
Mean caption length	8.2	7.4	–9.8%

Early warning sign

Caption length fell 8.2 → 7.4 words during RL — the CIDEr precision-over-recall bias appears immediately. This returns as a central theme in Sections 8 and 13.

6. Failed Directions (documented dead ends)

Experiment	Design	Outcome	Root cause
P2: DenseNet fine-tune	Unfreeze denseblock4, differential LR, CIDEr early stop	CIDEr 0.418 → 0.399, stop @ep2	8k images too few; BatchNorm stats destabilize
P3a: hard no-repeat-3gram	Hard-block trigram repeats at decode	Zero delta on all metrics	Soft 2-gram penalty already sufficient
P3b: focal loss retrain	$\gamma=2.0$, $\alpha=0.25$, full data, 3 attempts	Train loss falls but greedy BLEU-1 0.0038 (gibberish)	Focal down-weights easy tokens that carry caption structure at this scale

Conclusion of the DenseNet campaign

With decoder fixed (attention), tokenization fixed (BPE), and RL applied, the encoder was the last bottleneck — authorizing the CLIP pivot as the 'last resort'.

7. Generation 3 — CLIP ViT-B/16 Patch Features

7.1 Method

Swap ONLY the features: OpenAI CLIP ViT-B/16 (400M image-text pairs) → 196 patch tokens × 768 dims per image, precomputed for all 8,091 images as float16 memmap (2.4 GB). Decoder byte-identical to Gen 2 except `enc_dim 1920→768` (8,420,721 params). Implementation gotcha documented: `open_clip`'s `forward_intermediates` returns (1, 768, 14, 14) — `flatten(2).transpose(1,2)` to (196, 768); slicing off a class token corrupts features. Best epoch 25, val loss 3.7004 (37 epochs).

7.2 Results (full split)

Model	Decode	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L
Gen 2-RL (DenseNet)	beam-5 α 1.2	0.5971	0.3872	0.2456	0.1612	0.2566
Gen 3 CLIP (CE)	greedy	0.6520	0.4252	0.2590	0.1649	0.2774
Gen 3 CLIP (CE)	beam-5 α 1.2	0.6092	0.4344	0.2698	0.1755	0.2747

Key finding — the paper's central result

A pure feature swap beat EVERY previous model on EVERY metric. Even greedy CLIP (0.6520 BLEU-1) exceeded the entire previous beam-searched pipeline. Language-aligned pretraining, not decoder capacity, was the binding constraint at 8k-image scale.

8. Generation 3-RL — GRPO on the CLIP Decoder

8.1 Training dynamics

Identical GRPO recipe; images-per-step reduced 16 → 8 because CLIP's 196×768 attention quadruples memory vs DenseNet's 49×1920 (8 GB VRAM limit). Reward climbed monotonically 0.221 → 0.403. Validation CIDEr-D trajectory: 0.4859 → 0.4865 → 0.5142 → 0.5181 → 0.5211 → 0.5190 (best epoch 4, EMA served).

8.2 Full-split results

Model	Decode	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L
Gen 3 CLIP (CE)	beam-5 α 1.2	0.6092	0.4344	0.2698	0.1755	0.2747
Gen 3-RL greedy	greedy	0.6520	0.4252	0.2590	0.1649	0.2774
★ Gen 3-RL (final)	beam-5 α 1.2	0.6559	0.4344	0.2698	0.1727	0.2782

8.3 Diagnosing the length bias

A 30-image audit quantified the trade-off. GRPO word precision rose 70.7% vs CE 63.9% (+6.8 pts) — each emitted word is more likely correct — but captions shortened (~6 words) and audit BLEU-1 fell 0.519 → 0.477. A minimum-length decoding sweep:

Min length	BLEU-1	Word precision	Avg length
1 (none)	0.477	70.7%	6.0
8 (deployed) ★	0.533	70.4%	6.6
10	0.555	67.6%	8.2

Deployed fix

min_len=8 keeps RL's word-precision gain AND restores BLEU-1 above the CE model. Plus a trailing-function-word trimmer (captions never end on 'the', 'of', ...). Example: 'young boy' → 'young boy is in the grass'.

9. Follow-up Improvement Experiments (two more dead ends)

Three post-report experiments were proposed from multi-modal literature and tested empirically. Two failed; one was deprioritized on evidence.

9.1 Mixed reward $R = \text{CIDEr-D} + 0.5 \cdot \text{ROUGE-L}$ — REJECTED

Hypothesis: adding recall-friendly ROUGE-L to the reward fixes the length bias at the source (removing the min-len guard). Result: the opposite. ROUGE-L's LCS F-measure is itself precision-dominated on short captions, so it did not counteract CIDEr's bias; reward noise cost precision.

Metric	Pure CIDEr (rl)	Mixed (rl2)	Verdict
Best subset CIDEr-D	0.5211 (ep4)	0.5159 (ep3)	worse
Full-split BLEU-1 (beam-5)	0.6559	0.6514	worse
Word precision (30-img)	70.7%	68.0%	worse
Avg length after training	6.0	6.0	no effect

9.2 Diverse Beam Search — REJECTED

Hypothesis (Vijayakumar 2016): penalizing n-gram overlap between beams improves coverage of the 5 references. Result: monotonic degradation — Flickr8K's references are themselves similar, so diversity pushes beams toward low-probability phrasings matching nothing.

Decoding	BLEU-1	BLEU-4	Avg length
Standard beam (current) ★	0.5702	0.1510	7.5
DBS $\lambda=0.7$	0.5559	0.1450	7.0
DBS $\lambda=1.5$	0.5306	0.1432	6.8

9.3 Q-Former learnable queries — DEPRIORITIZED (analysis)

BLIP-2's Q-Former extracts visual semantics via K learnable queries, but it was trained on 129M images. At 8k images a from-scratch Q-Former adds random-init parameters precisely where data is scarcest; with only 196 tokens, cross-attention is not the bottleneck. Revisit only after Gen 4 (below) plateaus.

10. Generation 4 — Transformer Decoder: tested and rejected

Proposal from literature: replace the LSTM with a 4-layer pre-norm Transformer (8 heads, d=512, FFN 2048, dropout 0.3, causal self-attention + full cross-attention to all 196 patches). Trained with the identical recipe (AdamW wd 0.05, label smoothing, patience 12): best val loss 3.7941 @ epoch 41, early stop at 53. 23,360,368 parameters (+178% vs LSTM).

Metric	Gen 3-RL LSTM ★	Gen 4 Transformer	Δ
Full-split BLEU-1	0.6559	0.5546	−10.1 pts
Full-split BLEU-2	0.4344	0.3513	−8.3 pts
Full-split BLEU-4	0.1727	0.1348	−3.8 pts
Full-split ROUGE-L	0.2782	0.2414	−3.7 pts
Val loss	3.700	3.794	worse
Parameters	8.4M	23.4M	+178%

Why the transformer lost — three converging reasons

(1) Data scale: train loss fell to 2.19 while val stalled from epoch 12 — textbook overfitting; the LSTM's recurrence is itself a regularizer. (2) CLIP patches already carry structure; 196-token cross-attention adds capacity without new information. (3) Literature agrees: transformer decoders win at COCO+pretraining scale, not at 8k images.

Sample outputs (Gen 4 transformer — fluent but less accurate):

- "woman in white dress sitting in front of green shed"
- "young boy is jumping in the air in front of fountain"
- "two men are jumping into swimming pool on sunny day"

11. Deployment Engineering

11.1 Inference optimization (75×)

- Incremental state-carrying beam search replaced $O(T^2)$ prefix recomputation: 31,600 ms → ~420 ms per caption.
- Feature cache (16 images): repeat requests ~135 ms.
- fp16 encoder CUDA-only — CPU fp16 is EMULATED and cost 23 s per encode (measured); fp32 CPU is 0.5 s.
- Torch threads capped at 4; optional BLIP scorer (~1 GB) defaults off.

11.2 Infrastructure failure captured by telemetry

Second-resolution monitoring (per-process RSS/commit, system RAM, disk activity, queue depth → CSV) caught Next.js Turbopack inside OneDrive growing to 17.4 GB committed with system RAM at 15.91/15.91 GB and NVMe queue depth 122, followed by OOM cascade. Node's `--max-old-space-size` was ineffective (Rust-side memory). Fixes: `turbopackMemoryEviction 'full'`, explicit `turbopack.root`, `.next` junction off OneDrive (junctioning `node_modules` panics Turbopack — documented), standalone builds prod-only. Result: compiler footprint 7.7 GB → ~390-540 MB. ~11 GB of dead experiment artifacts purged after code-reference auditing.

12. Out-of-Distribution Analysis and Hybrid Routing

12.1 The failure mode

Outside Flickr8K's distribution the specialist hallucinates from its training prior: a cartoon becomes 'two people are standing in the of water'; a UI screenshot becomes 'black and white dog is standing'. Zero-shot probing proved the CLIP ENCODER represents these images correctly — the 8.4M-param decoder can only emit patterns from its 8k-image world.

12.2 Zero-shot routing fix

The already-loaded CLIP trunk classifies each request against six prompts (photo of people/animals outdoors; screenshot; food; cartoon; painting; document). In-domain → trained model; otherwise → locally stored BLIP. Routing reuses the encoder forward pass (near-zero cost), degrades gracefully if BLIP weights are missing, and is badged 'attention+blip (auto-routed)' in the UI. Verified: cartoon/screenshot route to BLIP and are captioned correctly; Flickr-style photos stay with the specialist.

12.3 Four-provider live-API evaluation (60 random val images)

Metric	CLIP+GRPO (ours)	Notebook TF	HF BLIP
BLEU-1	0.5739	0.4271	0.5033
BLEU-2	0.3712	0.2549	0.3284
BLEU-4	0.1324	0.0819	0.1288
ROUGE-L	0.2374	0.1760	0.2404
Word precision	69.4%	53.7%	68.1%
Mean length (words)	7.7	7.4	7.2
Median latency (ms)	416	7,009	2,511
P90 latency (ms)	443	9,713	3,178
Near-exact rate	5.0%	1.7%	3.3%

Interpretation

In-domain the specialist wins every fluency metric and is 6× faster than BLIP, 17× faster than TF. BLIP's tiny ROUGE-L edge reflects longer, recall-friendly sentences. The router composes both strengths.

13. Consolidated Results

Generation	Encoder	Decoder	BLEU-1	BLEU-4	ROUGE-L
Gen 1	DenseNet GAP (frozen)	LSTM-256 TF	0.5334	0.1218	0.2216
Gen 2	DenseNet 7×7 spatial	Bahdanau LSTM-512 + BPE	0.5649	0.1663	0.2548
Gen 2-RL	same	same + GRPO	0.5971	0.1612	0.2566
Gen 3	CLIP ViT-B/16 patches	Bahdanau LSTM-512	0.6092	0.1755	0.2747
★ Gen 3-RL (final)	same	same + GRPO + min-len 8	0.6559	0.1727	0.2782
Gen 4 (rejected)	same	Transformer 4L	0.5546	0.1348	0.2414

End-to-end progression: BLEU-1 +22.9%, BLEU-4 +41.8%, ROUGE-L +25.5% over baseline with a smaller decoder than the baseline generation. Rejected directions (mixed-reward GRPO, DBS, Gen-4 Transformer, Q-Former) are part of the record: theory predicted gains, measurement said otherwise.

14. Conclusions

- **Transfer dominates architecture at small scale.** A feature swap outperformed every decoder-side innovation combined (attention, RL, transformers).

- **RL on CIDEr trades recall for precision.** The length bias is systematic: handle at decode time (min-length guard worked) or reward time (ROUGE-L mixing failed empirically).
- **Specialists and generalists complement.** Zero-shot routing over an already-loaded encoder converts that complementarity into a product feature at near-zero cost.
- **Negative results have engineering value.** Five documented dead ends bound the search space for future work and prevented wasted compute.
- **Deployment is part of research.** Telemetry-driven diagnosis (65× inference, OOM root-cause) was required to make results usable at all.

Appendix A. Reproducibility

Artifact	Location
Gen 2 CE checkpoint	experiments/v2_attention/best.pt (ep24, val 3.8161)
Gen 2-RL checkpoint	experiments/v2_attention_rl/best.pt (ep5, EMA, CIDEr 0.4591)
Gen 3 CLIP checkpoint	experiments/v2_attention_clip/best.pt (ep25, val 3.7004)
Gen 3-RL checkpoint (served)	experiments/v2_attention_clip_rl/best.pt (ep4, EMA, CIDEr 0.5211)
Gen 3-RL2 mixed-reward (rejected)	experiments/v2_attention_clip_rl2/best.pt (ep3, CIDEr 0.5159)
Gen 4 Transformer (rejected)	experiments/v2_attention_tf/best.pt (ep41, val 3.7941)
CLIP features fp16	weights/clip_features_spatial.npy (8,091 × 196 × 768)
Provider evaluation JSON	%LOCALAPPDATA%/Temp/provider_eval.json
Telemetry monitor	%LOCALAPPDATA%/Temp/capmon.ps1 → captionai_telemetry.csv
Training scripts	backend/train_attention_v2*.py · eval_v2*.py
Serving stack	backend/inference/attention_provider.py (router, min-len, incremental beam)

Software: PyTorch 2.5.1+cu121, open_clip ViT-B-16/OpenAI, transformers BLIP-base, tokenizers BPE-6k, NLTK, FastAPI/Uvicorn, Next.js 16. Hardware: single 8 GB GPU, Windows 11, 16 GB RAM. All training seeded (42); audits seeded (7). Every number in this report traces to an artifact above.